

Brightcon - Brightway beginner course

Aalborg 2026

Learning objectives

Knowledge At the end of the course you will know about: python data structures, python scientific environment, Brightway2 data structures, Brightway2 functions, key concepts of statistical analysis for LCA (error propagation, statistical testing, variance-based sensitivity indicators).

Skills At the end of the course you will be able to: program in python at a basic level, use Brightway2 at a level comparable to other commercial LCA software, including importing foreground and background data into Brightway2, running calculations from a simple LCA to more complex simulations and comparative analyses, perform statistical analysis of LCA results, perform local and global sensitivity analysis of LCA results.

Competences At the end of the course you will be in a position to: apply LCA modelling in Brightway2 to your research questions, by choosing the appropriate data, code, and workflow organisation, including AI-supported workflow, that solve the case-specific challenges.

Required setup

What we need for the course is:

- A text editor, preferably *Visual Studio Code* (VSC).
- *Python* (the version is defined during brightway installation, if in doubt use python 3.10)
- Some scientific python packages like *scipy* and *numpy*
- Optionally, *Jupyter lab* which is a scientific python working environment
- *Brightway*, python-based software for LCA. You need the 2.5 version (also called *Brightway2.5* release)
- course notebooks, that are in a Github repository here:
<https://github.com/massimopizzol/advanced-lca-notebooks/tree/main/Course-material-bw25>
- A generative AI software, depending on what your organisation has license for.

!!IMPORTANT! Preparation before the course

- **Install the software.** Please make sure you have all the needed **software installed before** the course (we won't use lecture time to install software).

Option A. For the first time we are testing an AI assisted approach. Follow the instructions in the pdf file: [brightway25-AI-install-guide.pdf](#) If you have problems, write to Lotte Ansgaard Thomsen: lotteat@plan.aau.dk

Option B. Install brightway using the official informations from the website: [Brightway2 installation instructions](#).

- **Check the notebooks** linked above, and **watch the videos** here on <https://moutreach.science/2022/08/15/teaching-videos.html> under "*Videos for advanced LCA*". These videos use older versions of brightway but the content is essentially the same and you can follow-along with the notebooks.
- **Write down any question** you might have and bring it at the course.
- **Bring a case study.** This should be a simple, almost basic product system. 2-3 interconnected activities, ideally related to your project. We will use it for the exercises.

Plan for the day

(All exercises are in groups)

- Welcome and presentation round and AI setup (max 1h), introduction to LCA computation and general Brightway concepts (1h)
- Break
- Notebooks 1-2 (1h) + exercises (1h), Assisted by AI
- Lunch
- Notebooks 3-4 (1h) + exercises (1h), Assisted by AI
- Break
- Notebooks 5-6 (1h) + exercises (1h), Assisted by AI

(If we have time)

- Notebooks 7-8 + exercises, Assisted by AI

Contacts

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